Jiaxiong Tang

jxtang1@stu.ecnu.edu.cn
github.com/dickton

EDUCATION

East China Normal University 2024.09 - 2027.06
Software Engineering Master Shanghai, China
Research Interests: Trustworthy AI, Privacy & Ownership issues in AI

Dalian University of Technology 2020.09 - 2024.06
Cyber Engineering Bachelor Dalian, China

SKILLS

- Professional Skills: Architecture design, model selection, benchmarking, and ablation-driven iteration; Python, Pytorch; C++; Agentic workflow building; familiar with LLM fine-tuning & inference.
- Languages: Chinese (Native), English (Proficiency, IELTS 6.5), Japanese (JLPT N2)

PROFESSIONAL EXPERIENCE

Ubiquant (Top Quant in China) 2026.06 - Present
Quantitative Developer Intern Shanghai

- Develop and improve high-reliability trading infrastructure, including market connectivity, broker access, and automated validation pipelines.
- Develop the automated testing component of an end-to-end agentic software engineering system designed to automate the workflow from natural-language requirements through development, testing, and delivery.

RESEARCH EXPERIENCE

Robust Client-Server Watermarking for Split Federated Learning 2025.02 - 2025.06

- Designed and implemented a server–client joint watermarking framework to address model leakage and ambiguous ownership attribution in Split Federated Learning (SFL), enabling verifiable model ownership proof under the SFL architecture.
- Experimental results showed that the method maintained strong main-task performance, achieved over 95% detection accuracy with statistical significance ($p<0.03$), and preserved high watermark retention under attack scenarios.
- Outperformed baseline methods and showed good compatibility with large-scale SFL across multiple model architectures.
- Led code implementation, experimental design and evaluation, and paper writing, with deep involvement throughout the full research lifecycle from problem formulation to paper output. The paper is published on IoTJ

Sigil: Server-Enforced Watermarking in U-Shaped Split Federated Learning via Gradient Injection 2025.06 - 2025.11

- Given the limited model access in U-SFL, the Intellectual Property Rights of the server is under threat. Thus proposed an stealthy server-to-client watermarking embedded via gradient injection.
- Sigil achieves high detection rates, rubosness to SOTA anomalty detection methods and attacks in SFL.
- Highly engaged in project development, experiment evaluating and paper drafting. The paper is on the ***Arxiv***.

Self-adaptive Differential Privacy Framework for Split Federated Learning 2025.06 - 2026.03

- Addressed private data reconstruction risks in SFL, where existing dynamic differential privacy methods typically depend on full model knowledge or strong expert priors, leading to high deployment costs.
- Proposed an adaptive privacy-risk indicator using only local client-side information and developed a dynamic privacy budget allocation algorithm for client-limited settings, with its effectiveness validated through statistical experiments.
- Demonstrated through multiple SOTA privacy inference attacks that the method improved privacy protection by about 10% over expert-dependent baselines while preventing reconstruction of privacy data and maintaining comparable task accuracy.
- Led the project from problem formulation to paper output, covering method design, implementation, evaluation, and writing.

Stateful Backdoor Attacks on LLM Agents 2026.02 - 2026.06

- Proposed a stateful backdoor attack framework leveraging Agent Memory to enable stealthy, cross-session attack coordination.
- Fine-tuned Qwen3.5 and LLaMA3 with QLoRA in a simulated supply-chain poisoning scenario, allowing persistent attack states and covert malicious behaviors during authorized tool use.
- Validated on open-source LLMs, achieving high attack success rates despite existing safety alignment.
- Led method design, experimental studies, and paper writing across the full research workflow.

PROJECT EXPERIENCE

Small Target Detection in Extreme Noise via U-Net DNN 2025.08 - 2025.09
Technical Lead Collaborated with the National Space Science Center, CAS

- Led the end-to-end project lifecycle, independently handling requirements communication, algorithm development, system implementation, delivery testing, and iterative refinements.
- Developed an environmental simulation algorithm and implemented a U-Net network with 3D-convolution.
- Achieved 95%+ PD, <0.4% PFA and <1.0 px ALE under SNR=2dB (rough environment) while traditional method is invalid.
- Delivered the full system under a tight 6-week timeline with successful expert acceptance.